Using Data Science to Classify Galaxies
Case study by Diego McGrath

Imagine you've just joined a research team working with astronomers working with images from the Nancy Grace Roman Space Telescope, the successor to the Hubble Space Telescope. Modern telescopes like Nancy Grace capture tens of thousands of galaxy images, and scientists need help sorting them into meaningful categories. Traditionally, this work has been done by humans, but that's incredibly slow, and no longer a method that can keep up with our ever increasing number of telescopes in operation.

You've been tasked with building a machine learning system that can classify galaxies based on their visual appearance. These galaxies come in many forms, like spiral, smooth, merging, edge-on, and many more. Distinguishing between them isn't always straightforward, even for experts.

Your challenge is to explore whether a machine learning model, specifically a Convolutional Neural Network (CNN), can learn to recognize these patterns and classify galaxies with high accuracy.

The Deliverable:

You are tasked with classifying galaxies using a CNN with high accuracy, as well as seeing if your model performs equally well across all classes. You will need to investigate what kind of models can work efficiently, without sacrificing accuracy. To achieve these goals, you will work with a real dataset of over 17,000 galaxy images of multiple types, and apply both quantitative evaluation metrics and qualitative tools to see what the model is looking at.

At the end of this case study, your deliverable should show a functioning CNN that can efficiently analyze galaxy images and classify them correctly, along with analysis of how your model performed at classifying galaxies.

Some useful references and recommended models documentation:

Leung, H., & Bovy, J. (n.d.). *Galaxy10 DECaLS Dataset*. astroNN. https://astronn.readthedocs.io/en/latest/galaxy10.html#galaxy10-decals-dataset

Leung, H., & Bovy, J. (n.d.). *Galaxy10 DECaLS Dataset*. astroNN. https://astronn.readthedocs.io/en/latest/galaxy10sdss.html#tl-dr-for-beginners

Kazmierczak, J. (2026, March 9). *Galaxy Types*. NASA. https://science.nasa.gov/universe/galaxies/types/

Models and pre-trained weights — Torchvision main documentation. (n.d.). PyTorch documentation. Retrieved April 12, 2026, from https://docs.pytorch.org/vision/main/models.html

Explore NASA's website on the Nancy Grace Roman Space Telescope:
https://science.nasa.gov/mission/roman-space-telescope/